SnowPro Advanced: Data Engineer (DEA-C02) – Cheat Sheet

Data Movement & Ingestion

- COPY INTO for batch loads; Snowpipe for event-driven ingestion; Snowpipe Streaming for low-latency app ingestion.
- Snowpipe tracks loaded files automatically; transformation capability is limited.
- External stages preferred for cloud-native pipelines.

Trick Question: Snowpipe vs COPY INTO — choose COPY INTO when transformations and cost control matter.

Transformation, Streams & Tasks

- Streams track DML changes, not full tables.
- Task DAGs support dependencies but do not rollback upstream failures.
- Snowpark only when SQL cannot express the logic cleanly.

Trick Question: Streams do not delete rows; they advance offsets.

Performance Optimization

- Clustering improves partition pruning but increases cost if misused.
- Scale up warehouses for complexity; scale out for concurrency.
- Always analyze Query Profile before resizing warehouses.

Trick Question: Increasing warehouse size is rarely the first fix.

Storage & Data Protection

- Time Travel allows recovery; Fail-safe is last-resort only.
- Zero-copy cloning depends on Time Travel retention.
- Clones share storage until modified.

Trick Question: Fail-safe cannot be queried directly.

Security & Governance

- RBAC is role-centric; avoid direct user grants.
- Masking policies preferred over secure views.
- Row access policies dynamically filter data.

Trick Question: Secure views alone are insufficient for column-level security.

Data Sharing

- No data copying; consumer pays compute.
- Read-only access only.
- Secure and real-time.

Trick Question: Providers never pay compute costs.